

Rocksdb_dataset-2

Commit 21:

02b4197544f758bdf84d80fe9319238611848c48

Unable to provide test cases

Commit 22: 3570e4f5ffb29bc21b9afb388104a0a04f9af356

Non-C file modifications, therefore no performance regression testing is required.

Commit 23:

62a7ddb39c28d313d1d83350ccdf20dcad7bb6f6

Unable to provide test cases

Commit 24:

e48ccc28f4eebcc05b6333b129ee5908214d3259

The 1st symbolic execution

Testcase:for(int i = 0; i < 30000; i++){key = "key_missing" + std::to_string(distrib(gen));db->Put(rocksdb::WriteOptions(), key, value2);}

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  1991252.9259574472
4  The new version is executed by default
5  1993503.2048775507
6  The new version
7  2025508.8279387753
8  Comparison of New and old versions (along symbolic execution path)
9  -0.016912245214052258
10 Comparison of New and old versions (along the default execution path)
11 -0.015801275521368407
```

No performance regression

Commit 25: 159fa7741dec37c2e1f8c59c7fd9f5074984f613

The 1st symbolic execution

Testcase:for(int i = 0; i< 30000; i++){key = "key_missing" + std::to_string(distrib(gen));db->Put(rocksdb::WriteOptions(), key, value2);}

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  1975687.1462127657
4  The new version is executed by default
5  1979606.2653404255
6  The new version
7  1988375.654744681
8  Comparison of New and old versions (along symbolic execution path)
9  -0.006381343737355573
10 Comparison of New and old versions (along the default execution path)
11 -0.004410328291502579
```

No performance regression

The 2nd symbolic execution

Testcase:for(int i = 0; i< 1000; i++){db->SetOptions({{"write_buffer_size", "131072"}});}

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  835394.884368421
4  The new version is executed by default
5  866717.9429473685
6  The new version
7  857025.5095555555
```

```
8   Comparison of New and old versions (along symbolic execution path)
9   -0.025239184768667974
10  Comparison of New and old versions (along the default execution path)
11  0.011309387274644077
```

No performance regression

Commit 26:

601a6b59e895913d8d089341c9bc3ff5ad04e9eb

Non-C file modifications, therefore no performance regression testing is required.

Commit 27:

00a3eb6990504e2a6205c9d8321f6d6f95db0284

The 1st symbolic execution

```
Testcase:for(int i = 0; i< 30000; i++){key = "key_missing" + std::to_string(distrib(gen));db-
>Put(rocksdb::WriteOptions(), key, value2);}
```

No conditional branches, no need for symbolic execution.

代码块

```
1   end-to-end diff:
2   The new version executes along symbols
3   1988190.2163617017
4   The new version is executed by default
5   1976717.2120212764
6   The new version
7   1978342.5075918369
8   Comparison of New and old versions (along symbolic execution path)
9   0.004977757254911395
10  Comparison of New and old versions (along the default execution path)
11  -0.0008215440775919587
```

No performance regression

Commit 28:

d2db80caa28e21d0af8617b8a3d7cd83bbdb0443

Non-C file modifications, therefore no performance regression testing is required.

Commit 29:

3579d323c0d85492a20a8ef9da3e238cb2ac9253

Non-C file modifications, therefore no performance regression testing is required.

Commit 30:

f7d32c18a3416e1ebf5c2e6e058cb4d7c1f007a0

The 1st symbolic execution

```
Testcase:for (int i = 0; i < 100; i++){rocksdb::BackupEngine
*backup_engine;rocksdb::BackupEngine::Open(rocksdb::Env::Default(),
rocksdb::BackupEngineOptions("/tmp/test"), &backup_engine);backup_engine-
>CreateNewBackup(db);delete backup_engine;}
```

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  271220.2177446808
4  The new version is executed by default
5  271838.61327083333
6  The new version
7  267252.0918125
8  Comparison of New and old versions (along symbolic execution path)
9  0.014847876045680584
10 Comparison of New and old versions (along the default execution path)
11 0.017161779454101307
```

No performance regression

Commit 31:

631b6796b0142bcae3aa8be37426727bcad5b00d

The 1st symbolic execution

```
Testcase:for(int i =0;i < 100; i++){rocksdb::Slice start_key("key_exit:300"),
end_key("key_exit:500");db->DeleteRange(rocksdb::WriteOptions(), db-
>DefaultColumnFamily(), start_key, end_key);rocksdb::Slice
start_key1("key_exit:200");rocksdb::Slice
end_key2("key_exit:700");rocksdb::CompactRangeOptions comp_options;db-
>CompactRange(comp_options, &start_key1, &end_key2);}
```

Symbolic execution exception occurred, resulting in a timeout.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  1850623.6007399997
4  The new version is executed by default
5  1679186.0951020408
6  The new version
7  1727770.0530833334
8  Comparison of New and old versions (along symbolic execution path)
9  0.07110526510019376
10 Comparison of New and old versions (along the default execution path)
11 -0.028119458312517318
```

No performance regression

The 2nd symbolic execution

```
Testcase:for(int i = 0; i< 30000; i++){key = "key_missing" + std::to_string(distrib(gen));db-
>Put(rocksdb::WriteOptions(), key, value2);}
```

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  911179.8482631579
4  The new version is executed by default
5  903592.7835789475
6  The new version
7  900968.7975555556
8  Comparison of New and old versions (along symbolic execution path)
9  0.011333412139583731
```

```
10 Comparison of New and old versions (along the default execution path)
11 0.0029124049917279692
```

No performance regression

The 3rd symbolic execution

```
Testcase:for(int i =0;i < 100; i++){rocksdb::Slice start_key("key_exit:300"),
end_key("key_exit:500");db->DeleteRange(rocksdb::WriteOptions(), db-
>DefaultColumnFamily(), start_key, end_key);rocksdb::Slice
start_key1("key_exit:200");rocksdb::Slice
end_key2("key_exit:700");rocksdb::CompactRangeOptions comp_options;db-
>CompactRange(comp_options, &start_key1, &end_key2);}
```

Symbolic execution completed successfully.

代码块

```
1 end-to-end diff:
2 The new version executes along symbols
3 17202505.11622222
4 The new version is executed by default
5 16966468.663105264
6 The new version
7 17026605.23411111
8 Comparison of New and old versions (along symbolic execution path)
9 0.010330883913295404
10 Comparison of New and old versions (along the default execution path)
11 -0.003531917853205973
```

No performance regression

The 4th symbolic execution

```
Testcase:for(int i =0;i < 100; i++){rocksdb::Slice start_key("key_exit:300"),
end_key("key_exit:500");db->DeleteRange(rocksdb::WriteOptions(), db-
>DefaultColumnFamily(), start_key, end_key);rocksdb::Slice
start_key1("key_exit:200");rocksdb::Slice
end_key2("key_exit:700");rocksdb::CompactRangeOptions comp_options;db-
>CompactRange(comp_options, &start_key1, &end_key2);}
```

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  1692515.9665957442
4  The new version is executed by default
5  1705901.8205319152
6  The new version
7  1700619.843625
8  Comparison of New and old versions (along symbolic execution path)
9  -0.004765249011784847
10 Comparison of New and old versions (along the default execution path)
11 0.0031059127803990936
```

No performance regression

The 5th symbolic execution

Testcase:for(int i = 0; i< 30000; i++){key = "key_missing" + std::to_string(distrib(gen));db->Put(rocksdb::WriteOptions(), key, value2);}

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  1990669.5381836733
4  The new version is executed by default
5  1983635.2958541668
6  The new version
7  1980199.7422500001
8  Comparison of New and old versions (along symbolic execution path)
9  0.005287242347469888
10 Comparison of New and old versions (along the default execution path)
11 0.00173495306097913
```

No performance regression

The 6th symbolic execution

```
Testcase:for(int i =0;i < 100; i++){rocksdb::Slice start_key("key_exit:300"),
end_key("key_exit:500");db->DeleteRange(rocksdb::WriteOptions(), db-
>DefaultColumnFamily(), start_key, end_key);rocksdb::Slice
start_key1("key_exit:200");rocksdb::Slice
end_key2("key_exit:700");rocksdb::CompactRangeOptions comp_options;db-
>CompactRange(comp_options, &start_key1, &end_key2);}
```

Symbolic execution exception occurred, resulting in a timeout.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  1689900.3843125
4  The new version is executed by default
5  1685387.38400000003
6  The new version
7  1709734.2442708332
8  Comparison of New and old versions (along symbolic execution path)
9  -0.011600551386740186
10 Comparison of New and old versions (along the default execution path)
11 -0.014240143082129316
```

No performance regression

Commit 32:

09d7f6a5c29c367b0662ee43ad1df993b5dfc302

The 1st symbolic execution

```
Testcase:for(int i = 0; i< 30000; i++){key = "key_missing" + std::to_string(distrib(gen));db-
>Put(rocksdb::WriteOptions(), key, value2);}
```

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  1982729.206744681
4  The new version is executed by default
5  1983652.9968541667
```

```
6 The new version
7 1976413.112836735
8 Comparison of New and old versions (along symbolic execution path)
9 0.0031957356824457448
10 Comparison of New and old versions (along the default execution path)
11 0.0036631430799607865
```

No performance regression

Commit 33:

c2de7832bc888edcbde03ffeb6b437836b8a5f71

Non-C file modifications, therefore no performance regression testing is required.

Commit 34:

9b1d0c02e9bcf4377b0dc638441c11c4aa10e8b7

Non-C file modifications, therefore no performance regression testing is required.

Commit 35:

b341dc8b05bd0e2499e22886733f25ad011d2a50

Unable to provide test cases

Commit 36:

d4bd67fb0972bcf654f9d63bdabc39001a8208b7

Non-C file modifications, therefore no performance regression testing is required.

Commit 37:

b5e41625828a8ebec3e46f85664f862322c726a2

The 1st symbolic execution

```
Testcase:for (int i = 0; i < 100; i++){rocksdb::BackupEngine
*backup_engine;rocksdb::BackupEngine::Open(rocksdb::Env::Default(),
rocksdb::BackupEngineOptions("/tmp/test"), &backup_engine);backup_engine-
>CreateNewBackup(db);delete backup_engine;}
```

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  272259.1067659575
4  The new version is executed by default
5  270664.1491020408
6  The new version
7  267464.73922000005
8  Comparison of New and old versions (along symbolic execution path)
9  0.01792523216308484
10 Comparison of New and old versions (along the default execution path)
11 0.011961987555335713
```

No performance regression

Commit 38:

44b741e9ccc2a99334806c08e8b09c3ad8010f63

The 1st symbolic execution

Testcase:for(int i = 0; i< 30000; i++){key = "key_missing" + std::to_string(distrib(gen));db->Put(rocksdb::WriteOptions(), key, value2);}

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  2045280.3994583332
4  The new version is executed by default
5  2040620.4125625
6  The new version
7  2036317.3625510205
8  Comparison of New and old versions (along symbolic execution path)
9  0.004401591359062098
10 Comparison of New and old versions (along the default execution path)
11 0.0021131529351047875
```

No performance regression

Commit 39:

6c6defe3b8c8eafc01cdb5a8b5834ae0c2149298

The 1st symbolic execution

```
Testcase:for (int i = 0; i < 100; i++){rocksdb::BackupEngine
*backup_engine;rocksdb::BackupEngine::Open(rocksdb::Env::Default(),
rocksdb::BackupEngineOptions("/tmp/test"), &backup_engine);backup_engine-
>CreateNewBackup(db);delete backup_engine;}
```

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  270202.73156
4  The new version is executed by default
5  266637.22347826086
6  The new version
7  269473.95407999994
8  Comparison of New and old versions (along symbolic execution path)
9  0.002704444971270554
10 Comparison of New and old versions (along the default execution path)
11 -0.010526919425010283
```

No performance regression

The 2nd symbolic execution

```
Testcase:for(int i = 0; i < 30000; i++){key = "key_missing" + std::to_string(distrib(gen));db-
>Put(rocksdb::WriteOptions(), key, value2);}
```

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  1990729.4886734695
4  The new version is executed by default
5  1976343.7058125
6  The new version
```

```
7 1984188.714591837
8 Comparison of New and old versions (along symbolic execution path)
9 0.0032964475775571493
10 Comparison of New and old versions (along the default execution path)
11 -0.003953761414750668
```

No performance regression

The 3rd symbolic execution

```
Testcase:for (int i = 0; i < 100; i++){rocksdb::BackupEngine
*backup_engine;rocksdb::BackupEngine::Open(rocksdb::Env::Default(),
rocksdb::BackupEngineOptions("/tmp/test"), &backup_engine);backup_engine-
>CreateNewBackup(db);delete backup_engine;}
```

No conditional branches, no need for symbolic execution.

代码块

```
1 end-to-end diff:
2 The new version executes along symbols
3 272833.8471914893
4 The new version is executed by default
5 269376.55946938775
6 The new version
7 268290.7971875
8 Comparison of New and old versions (along symbolic execution path)
9 0.016933305397032736
10 Comparison of New and old versions (along the default execution path)
11 0.00404696058631095
```

No performance regression

Commit 40:

3040868e5fd8471c48878baf7ce90c6e1667e67a

Non-C file modifications, therefore no performance regression testing is required.